

Problem Set B — Data and Simulation (Weeks 3 – 4)

Programming and Numerical Methods for Economics (ECNM)

The University of Edinburgh · School of Economics

Instructions. This problem set covers material from Weeks 3 and 4 : pandas DataFrame indexing, conditional subsetting, dummy variables, missing values, random number generation, distributions, Monte Carlo integration, AR(1) processes, and Markov chains.

```
In [ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
np.random.seed(42)
```

Question 1 — Building and exploring a DataFrame

The table below shows data on six European economies:

Country	GDP (bn €)	Population (m)	Unemployment (%)	Year
Germany	3.5	82.4	3.5	2013
France	2.7	64.6	6.5	2013
Italy	2.1	60.4	11.2	2013
Spain	1.8	45.3	24.5	2013
Netherlands	0.5	16.5	5.2	2013
Greece	0.2	11.1	26.5	2013

- (a) Create a DataFrame `eu` from this data (use any method you prefer).
- (b) Add a new column `gdp_pc` equal to GDP per capita (GDP / Population). Note the units.
- (c) Select only the countries where unemployment is **above** 10%. Which countries appear?
- (d) Select countries with GDP per capita **above the median** AND unemployment **below** 10%. Only the `Country` and `gdp_pc` columns.

```
In [ ]: # Your answer here
```

Question — Dummies, missing values, and data cleaning

The DataFrame below records households in a survey:

```
survey = pd.DataFrame({
    'hh_id': [1, 2, 3, 4, 5, 6, 7],
    'income': [2500, 4100, np.nan, 1800, 3200, np.nan, 5500],
    'region': ['urban', 'rural', 'urban', 'rural', 'urban', 'rural', 'urban'],
    'children': [2, 0, 1, np.nan, 3, 1, np.nan]
})
```

- (a) Create this DataFrame. How many missing values does each variable have?
- (b) Drop all rows where `income` is missing. How many rows survive?
- (c) On the surviving data, fill missing `children` with the **median** number of children (compute non-missing values).
- (d) Create a dummy variable `urban` equal to `1` if region is `'urban'`, otherwise `0`.
- (e) Compute the mean income separately for urban and rural households (use the cleaned data) group has a higher average?

In []: `# Your answer here`

Question — Random numbers and distributions

- (a) Set the seed to `12345`. Draw `1000` observations from a Normal distribution with $\mu = 0$ and $\sigma = 1$. Store them in `x`.
- (b) Draw `1000` observations from a **log-normal** distribution by computing `z = np.exp(x)`. Plot histograms of both `x` and `z` side by side (use `plt.subplots(1, 2)`). 1 panel.
- (c) Create your own discrete distribution with support `[0, 1, 2, 3]` and probabilities `[0.5, 0.2, 0.1, 0.2]`. Draw `1000` samples from it. Verify empirically that the sample mean is close to the theoretical mean $E[X] = \sum_i x_i p_i$.

In []: `# Your answer here`

Question — Monte Carlo integration

Suppose $X \sim N(0, 1)$ and we want to compute $E[g(X)]$ where $g(x) = e^{-x^2/2}$.

- (a) Write a function `g(x)` implementing the expression above.
- (b) Estimate $E[g(X)]$ by Monte Carlo with $N = \quad \quad \quad$ draws. Print the res
- (c) Investigate convergence: compute the Monte Carlo estimate for $N = \quad \quad \quad$, $\quad \quad \quad$, $\quad \quad \quad$, $\quad \quad \quad$ \$. Store the estimates in a list them against N (use a log scale on the x-axis). Does the estimate stabilise?

In []: `# Your answer here`

Question — AR() processes

Consider the AR() process: $y_{t+1} = a + \rho y_t + \varepsilon_t$, where $\varepsilon_t \sim N(0, \sigma^2)$.

- (a) Write a function `simulate_ar1(T, rho, a=0, sigma=1, y0=0)` that returns a NumF of length T containing the simulated path.
- (b) Set `rho=0.9`, `a=1`, `sigma=0.5`, `T=200`. Simulate and plot the process. Add a horizon at the **unconditional mean** $\mu = a/(1-\rho)$. Does the process fluctuate around this v
- (c) Now set `rho=1.01` (keeping the other parameters). Simulate periods and plo happens? Is this process stationary? Explain in one sentence.

In []: `# Your answer here`

Question — Markov chains

An economy has three states of the business cycle: **Recession**, **Normal**, and **Boom**. The quarterly transition matrix is:

$$P = \begin{pmatrix} . & . & . \\ . & . & . \\ . & . & . \end{pmatrix}$$

where rows are current states (Recession, Normal, Boom) and columns are next-period states.

- (a) Use `quantecon.MarkovChain` to create this Markov chain with state labels `['Recessi', 'Normal', 'Boom']`.
- (b) Simulate periods starting from `'Normal'`. Compute the fraction of time each state.
- (c) Compute the **stationary distribution** using `mc.stationary_distributions`. Compare your simulation fractions from (b).

(d) Suppose GDP growth in each state is: Recession = – %, Normal = %, Boom = %
the stationary distribution, what is the **long-run average** GDP growth rate for this economy?

In []: # Your answer here

End of Problem Set B.